

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards (BL2,BL6)

Assessment Tool Weightage: 30%

QUESTIONS: 5

Total Marks:25

Annexure A

Comparative Analysis

1. Star topology is generally more expensive than bus topology because it requires a central device such as a switch or hub and more cabling to connect each node individually. In contrast, bus topology uses a single backbone cable, making it low-cost and easier to install. However, in terms of reliability, star topology is superior because failure of one node does not affect others, whereas a failure in the main bus cable can bring down the entire network. Performance is also better in star topology since data transmission is managed through a central device, reducing collisions, while bus topology suffers from performance degradation as more devices are added due to increased data traffic on the shared medium.
2. Mesh topology offers very high fault tolerance because every node is connected to multiple other nodes, providing multiple paths for data transmission. If one link fails, data can still be routed through alternate paths without disrupting communication. In contrast, star topology has moderate fault tolerance; while failure of individual nodes does not affect the network, failure of the central hub or switch can cause the entire network to stop functioning. Therefore, mesh topology is more robust and reliable in critical environments compared to star topology.
3. Bus topology is simple and easy to install because it uses a single backbone cable with minimal connections, making it suitable for small networks. On the other hand, mesh topology is highly complex to install as each node must be connected to every other node, requiring a large number of cables and ports. This complexity increases significantly as the number of devices grows, making mesh topology time-consuming and costly to implement compared to the straightforward setup of bus topology.
4. Hybrid topology is highly scalable because it combines two or more different topologies (such as star, mesh, or bus), allowing the network to expand flexibly according to organizational needs. New segments can be added without affecting the existing network structure. In comparison, star topology is also scalable but to a limited extent, as it depends on the capacity of the central device (switch or hub). Once the central device reaches its port limit, expansion requires additional hardware. Thus, hybrid topology provides greater scalability than star topology.
5. In terms of maintenance, star topology is relatively easy to manage because issues can be quickly identified and isolated to a specific node or cable. Mesh topology, while highly reliable, is difficult to maintain due to its complex structure and large number of connections, making troubleshooting time-consuming. Bus topology is harder to maintain than star because faults in the backbone cable can disrupt the entire network and are difficult to locate. Hybrid topology has moderate maintenance complexity; while it benefits from flexibility, managing multiple combined topologies requires careful monitoring and skilled administration.

RUBRICS FOR CALCULATION

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding of Concepts	15%	Demonstrates complete and indepth understanding of all concepts being compared	Good understanding with minor gaps	Basic understanding; some concepts unclear	Poor understanding of concepts
Comparison Depth	20%	Provides detailed, insightful, and meaningful comparisons across multiple aspects	Adequate comparison with relevant points	Limited comparison; lacks depth	Minimal or incorrect comparison
Criteria Selection	10%	Selects highly relevant and appropriate comparison parameters	Mostly relevant criteria	Some irrelevant or missing criteria	Poor or no criteria selection
Analytical Skills	15%	Strong analysis with logical reasoning and clear justification	Good analysis with some justification	Basic analysis with weak reasoning	No clear analysis
Organization & Structure	10%	Well-organized, clear, and logically structured	Generally organized	Somewhat disorganized	Poorly structured
Use of Examples / Evidence	10%	Uses strong, relevant examples or data to support comparison	Uses some examples	Limited examples	No supporting examples
Clarity of Presentation	10%	Very clear, concise, and easy to understand	Mostly clear	Somewhat unclear	Difficult to understand
Conclusion & Insights	10%	Insightful conclusion with strong justification	Clear conclusion	Basic conclusion	No or weak conclusion

Star Topology vs Bus Topology

Star topology and Bus topology are two common network topologies. They differ in terms of cost, reliability, performance, installation, and maintenance.

1. Cost

Star topology is generally more expensive than bus topology. It requires a central device such as a switch or hub and separate cables to connect each device to the central device.

Bus topology is cheaper because all devices share a single backbone cable. It requires less cable and fewer networking devices.

2. Installation

Star topology requires more cabling because every node must have an individual connection to the central switch or hub. Therefore, the initial installation cost and effort are higher.

Bus topology is simple and easy to install because devices are connected to one common backbone cable.

3. Reliability

Star topology provides better reliability. If one computer or its connecting cable fails, only that particular device is affected. The remaining devices can continue to communicate.

In bus topology, failure of the main backbone cable can affect the entire network, making it less reliable.

4. Performance

Star topology generally provides better performance because communication is managed through the central switch. A switch can forward data to the appropriate destination device, reducing unnecessary traffic and collisions.

In bus topology, all devices share the same communication medium. When more devices are added, network traffic increases and performance can decrease.

5. Fault Detection

Fault detection is easier in star topology because every device has a separate connection to the central device. If a connection fails, it is easier to identify the affected cable or device.

In bus topology, finding a fault in the shared backbone cable can be more difficult because the failure can affect multiple devices.

6. Network Expansion

Star topology is easier to expand. A new computer can be connected to an available port on the switch without disturbing the existing devices.

Expanding a bus network can be more difficult because additional devices must be connected to the shared backbone.

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2. Mesh Topology vs Star Topology – Fault Tolerance

Mesh topology and **Star topology** are two important network topologies. They differ significantly in terms of fault tolerance, reliability, cost, maintenance, and performance.

1. Fault Tolerance in Mesh Topology

Mesh topology provides **very high fault tolerance** because devices have multiple connections to other devices. This creates multiple paths for data transmission.

If one network link fails, data can use another available path. Therefore, communication can continue without major interruption.

2. Fault Tolerance in Star Topology

Star topology has **moderate fault tolerance**. Each device is connected independently to a central switch or hub.

If one computer or its connecting cable fails, only that particular device is affected. Other devices can continue communicating.

However, the central switch is a **single point of failure**. If the switch or hub fails, communication between all connected devices is interrupted.

3. Reliability

Mesh topology is more reliable because it provides multiple communication paths. A single link failure normally does not stop communication.

Star topology is also reliable for individual device failures, but its overall reliability depends heavily on the central switch.

4. Alternative Paths

One of the main advantages of mesh topology is the availability of **alternative paths**.

For example, if a direct connection between two devices becomes unavailable, data can travel through another connected device.

Star topology does not provide alternative paths between the end devices. All communication depends on the central switch.

5. Network Failure

In a mesh network, failure of one or even several links may not cause complete network failure.

In a star network, failure of a single cable normally affects only one device. But failure of the central switch can affect the **entire network**.

6. Cost

Mesh topology is expensive because it requires many connections, cables, and network interfaces.

Star topology is generally less expensive than a full mesh because each device requires only one connection to the central switch.

7. Installation Complexity

Mesh topology is difficult to install and configure because many connections are required between devices.

Star topology is easier to install because each device is connected directly to a central switch.

8. Maintenance

Maintaining a mesh network can be complex because there are many links to monitor and troubleshoot.

Star topology is easier to maintain because a faulty device or cable can be identified and replaced without affecting the other devices.

3. Bus Topology vs Mesh Topology – Installation Complexity

Bus topology and **Mesh topology** are two different network topologies with significant differences in installation, cost, cabling, maintenance, reliability, and scalability.

1. Bus Topology

In bus topology, all computers and network devices are connected to a **single main backbone cable**. The backbone acts as the common communication medium for all devices.

Bus topology requires fewer cables and connections, making it **simple and easy to install**.

2. Simple Installation

Bus topology is straightforward to install because devices are connected to the same backbone cable. It does not require a separate connection between every pair of devices.

This makes it suitable for **small networks** where only a few devices are connected.

3. Low Cost

Bus topology is relatively inexpensive because it requires:

- Less cabling
- Fewer connectors
- No central switch or hub
- Less installation equipment

Therefore, it is useful when the primary requirement is **low installation cost**.

4. Mesh Topology

In mesh topology, devices are connected to multiple other devices. In a **full mesh**, every device has a direct connection to every other device.

This provides multiple paths for data transmission but makes the network much more complicated to install.

5. Large Number of Connections

The number of connections required in a full mesh network can be calculated using:

$$\text{Number of links} = \frac{n(n-1)}{2}$$

where n is the number of devices.

For example, if there are 5 devices:

$$\frac{5(5-1)}{2}$$

$$\begin{aligned}
 &= \frac{5 \times 4}{2} \\
 &= 10 \text{ links}
 \end{aligned}$$

Thus, 5 devices require **10 direct connections** in a full mesh.

6. High Installation Cost

Mesh topology requires a large amount of cabling, ports, connectors, and network interfaces. As the number of devices increases, the number of connections increases rapidly.

Therefore, mesh topology is **more expensive** to install than bus topology.

7. Installation Time

Bus topology requires relatively less installation time because there are fewer connections.

Mesh topology takes considerably more time because each device needs multiple connections, and every connection must be properly configured and tested.

8. Maintenance Complexity

Bus topology is easier to maintain because there are fewer components. However, troubleshooting the backbone can sometimes be difficult because a backbone failure can affect the entire network.

Mesh topology is more difficult to maintain because there are many links and connections to monitor. However, its multiple paths provide better reliability when individual links fail.

4. Hybrid Topology vs Star Topology – Scalability

Hybrid topology and **Star topology** are commonly used network topologies. Both can support network expansion, but hybrid topology provides greater flexibility and scalability because it combines two or more different topologies.

1. Hybrid Topology

Hybrid topology is a combination of **two or more different network topologies**, such as Star, Bus, Ring, or Mesh.

It allows an organization to design different parts of a network according to their specific requirements.

2. High Scalability

The major advantage of hybrid topology is its **high scalability**. New network segments can be added without completely changing the existing network structure.

For example, an organization can use a star topology for one department and connect it with another department using a different topology.

3. Flexible Expansion

Hybrid topology provides flexibility because different sections of the network can use different technologies.

If an organization expands, a new network segment can be added to the existing hybrid network according to the requirements of that department or location.

4. Star Topology

In star topology, all devices are connected to a **central switch or hub**. It is also relatively easy to expand because a new device can be connected to an available port on the central device.

However, its scalability depends on the **capacity and number of ports** available on the switch or hub.

5. Limitation of Star Topology

If the central switch has no free ports, additional hardware is required.

For example, if a switch supports 24 devices and all 24 ports are occupied, adding another device requires:

- A larger switch, or
- An additional switch connected to the existing network.

Therefore, expansion can increase the hardware and management requirements.

6. Scalability of Hybrid Topology

Hybrid topology overcomes some of these limitations by allowing multiple network segments to be combined.

A large organization can have different departments with their own network structures and connect them together.

This makes hybrid topology suitable for **large organizations, universities, hospitals, and multinational companies**.

7. Cost

Hybrid topology can be more expensive because it may require different types of networking devices, cables, switches, routers, and other infrastructure.

Star topology is generally less expensive and easier to implement for a small or medium-sized network.

Maintenance of Star, Mesh, Bus and Hybrid Topologies

Network maintenance refers to the process of **monitoring, identifying faults, repairing devices, and ensuring that the network operates properly**. The maintenance difficulty differs between Star, Mesh, Bus, and Hybrid topologies.

1. Star Topology

Star topology is **relatively easy to maintain** because every device has a separate connection to a central switch or hub.

If one computer or cable fails, the problem can usually be identified quickly. The other devices can continue working normally.

For example, if one network cable is damaged, only the device connected through that cable is affected.

Advantages in maintenance:

- Easy fault detection
- Easy troubleshooting
- Individual devices can be repaired separately
- Failure of one node does not normally affect other nodes
- Network expansion and replacement are simple

Therefore, star topology is widely preferred because of its **simple maintenance and management**.

2. Mesh Topology

Mesh topology is **difficult to maintain** because it contains a large number of connections between devices.

When a problem occurs, administrators need to check several links and devices to identify the faulty connection.

Although troubleshooting is complex, mesh topology provides excellent reliability because alternative paths are available.

Maintenance characteristics:

- Large number of connections
- Difficult fault identification
- Higher maintenance cost
- Requires skilled network administrators
- Multiple alternative paths improve reliability

Thus, mesh topology provides high reliability but requires **more effort and expertise for maintenance**.

3. Bus Topology

Bus topology is harder to maintain than star topology because all devices share a common **backbone cable**.

If the backbone cable is damaged, communication can be affected across the entire network. Locating the exact point of failure can be difficult.

Maintenance problems:

- Backbone failure can affect the entire network
- Fault location can be difficult
- Shared cable requires careful troubleshooting
- Adding or removing devices can affect the network
- Performance problems can occur as network traffic increases

Therefore, bus topology has **higher maintenance difficulty** compared with star topology.

4. Hybrid Topology

Hybrid topology combines two or more different topologies, such as star, bus, ring, or mesh.

It provides flexibility, but maintenance can be more complex because different sections may use different network structures and technologies.